Trails Revisited: Associative Indexing in Personal Knowledge Tools Eighty Years After the Memex

Elisa Marchetti

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Abstract

Bush's memex proposed that a reader could tie two items together and later replay the chain of associations as a trail. Contemporary personal knowledge tools implement links and backlinks but rarely trails: ordered, named, shareable paths through material. We analyze 18 tools and find that 16 support bidirectional links, 5 support ordered sequences, and none preserves the reason a connection was made. We argue the reason, not the link, was the memex's central idea.

1 Introduction

The memex is often cited as the ancestor of hypertext. What Bush actually emphasized was the trail: a named sequence of items, built while thinking, that could be handed to someone else. Links are a mechanism; trails are a record of thought.

2 Tools analyzed

We examined 18 personal knowledge tools available in 2024: note apps with backlinks, outliners, canvas tools and reference managers.

Capability	Tools (of 18)
Bidirectional links	16
Graph view	12
Ordered, named sequences	5
Sharing a sequence with others	3
Recording why a link was made	0

3 What is missing

Every tool lets a user say that two notes are related. None records why, at the time the connection is made, in a form the tool can use. Users who want this write it into the note body, where it is invisible to the link structure.

Five tools support sequences, but sequences are authored after the fact as tables of contents, not accumulated during reading.

4 Design proposals

- Capture the context of a link at creation: what was open, what was just read, what the user typed.
- Treat trails as first-class objects that can be replayed and forked.
- Let a trail survive export.

5 Conclusion

The question "where did this idea come from?" is answerable only if the tool keeps the path, not just the endpoints.

References

[1] Bush, V. As we may think. The Atlantic Monthly 176, 1 (1945), 101-108.

[2] Nelson, T. H. Complex information processing: a file structure for the complex, the changing and the indeterminate. In Proceedings of the 20th ACM National Conference (1965), 84-100.